ADDENDUM TO *A5K* · REV. K

Where This Design Sits

AAA, Hombre, the shipped Amiga eras, and a 2026 scientific workstation — compared honestly

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SCOPE AND GROUND RULES

This addendum does not restate the design; it places it. It compares the A5K concept against the two Amiga architectures that were designed but never built — **AAA** (1994) and **Hombre** (1995) — against the three that shipped, and against a current top-end scientific workstation.

One asymmetry governs everything below. **OCS, ECS, AGA and the workstation are real, shipping hardware. AAA and Hombre were cancelled before volume silicon, and the A5K is a paper design with a bill of materials.** Every comparison involving the Amigas is therefore design-target against design-target, and every number is an estimate derived from public specifications, datasheets and the BOM in Appendix C of the main document. Where a figure is a range, the range *is* the answer.

1. Why the A5K is better than AAA — and by how far

AAA (1993–94) was a **2D evolution of the Amiga chipset**: 32-bit paths, 16-bit chunky as well as planar, more bitplanes, 8-channel 16-bit audio, a 64-bit blitter, a wider Copper, and a display ceiling around 1280×1024. It was the right architectural answer to AGA's limits, and it was cancelled because Commodore ran out of money before it could tape out.

The A5K beats it in five ways and loses in one.

Table A1. The A5K against the AAA target specification.

Axis	AAA (target, ~1995 silicon)	A5K (2026, FPGA)	Ratio
Configurable logic	~50–150k gates across 3–4 chips	215k logic cells ≈ 2–4M gate-equivalents	~20–60×
Clock domain	~14–28 MHz chip bus	100–200 MHz core; 148.5 MHz pixel	~5–10×
Chip RAM	~2–16 MB (never fixed)	512 MB – 2 GB DDR3/LPDDR4	~100–1000×
Memory bandwidth	~50–100 MB/s	6.4–12.8 GB/s	~100×
Host link	Zorro III, ~37 MB/s	PCIe Gen1/2 ×4, 1–2 GB/s	~30–50×
Scanout	1280 × 1024	4K (3840 × 2160)	~6× pixels
Native 68k software	Yes, by construction	No — AROS or emulation	AAA wins

By how far: roughly **1.5 orders of magnitude in silicon, and 2–3 in memory and bandwidth**. That is not a new class of machine — it is the AAA concept executed on a 2026 FPGA with twenty-five extra years of process, memory and I/O. The real difference

is not the ratio but the fact that **one shipped and the other didn't.** AAA needed a fab, a CPU, an OS team, a factory and a market; the A5K needs a $173 BOM and a laptop.

The one place AAA is strictly better: it would have run 1994 Amiga software natively. The A5K cannot, and should not pretend otherwise.

2. Why it is better than Hombre — and where Hombre was ahead

Hombre (1995) was a different animal: a **PA-RISC CPU plus a fixed-function 3D chipset**, aimed at a CD32 successor — a games console rather than a computer. It was cancelled in 1995–96.

Against Hombre, the A5K wins on:

- **Openness.** PA-RISC is dead; the design would have been closed, single-vendor and unmaintainable. The A5K is RISC-V or x86 plus LiteX and AROS, all readable.
- **Memory and I/O.** Hombre would have had a few megabytes and a cartridge or CD bus. The A5K has 2 GB and PCIe.
- **Survivability.** Hombre depended on winning a console war against PlayStation. The A5K depends on nobody.
- **Tools.** Yosys, nextpnr and Verilator, versus 1995 proprietary everything.

But be honest about the 3D: **Hombre had a 3D pipeline; the A5K has none.** Its likely target was 10^5–10^6 polygons per second — three to four orders below a 2026 GPU — so the correct comparison is not "ours is faster" but "ours declines the contest and borrows a GPU as a plane source" (§9.6 of the main document). Hombre tried to win a race it was already losing; the A5K refuses to enter it.

By how far: the same 1.5–3 orders in memory and logic, plus qualitative wins in openness and cost — but *minus* the one thing Hombre actually had, which is 3D silicon.

3. The three Amiga eras, side by side

Table A2. Shipped Amigas, cancelled Amigas, and the paper design.

	OCS (1985)	ECS (1990)	AGA (1992)	AAA (1994, †)	Hombre (1995, †)	A5K (2026, paper)
CPU	68000 @ 7.1 MHz	68000 / 020	020 / 030 / 040	040 / 060	PA-RISC	RISC-V or x86 host
Chip RAM	256–512 KB	1–2 MB	2 MB	~2–16 MB	a few MB	0.5–2 GB
Planes / colour	6 / 4,096	6 / 4,096	8 / 262k (HAM8)	16-bit chunky	3D	8-planar + ARGB planes
Blitter	16-bit	16-bit	16-bit	64-bit	3D	64-bit equivalent
Audio	4 × 8-bit	4 × 8-bit	4 × 8-bit	8 × 16-bit	CD-class	N × 24-bit I²S DMA
Display	640 × 512	640 × 512	1280 × 512	~1280 × 1024	TV	4K
Bus	custom	custom	custom	custom	custom	PCIe / AXI
Software	AmigaOS	AmigaOS	AmigaOS	—	—	AROS / Linux
Exists?	**yes**	**yes**	**yes**	no	no	no

The A5K is not the next Amiga era. It is **AAA's architecture, Hombre's ambition minus the 3D, and neither one's market** — instantiated on hardware that exists.

4. Against a cutting-edge scientific workstation

Take a realistic top-end mathematics-lab node: dual 128-core EPYC or Xeon 6, 2–3 TB of DDR5, eight H100/B200-class accelerators, 100+ TB of NVMe and 400 Gb InfiniBand — or, for a single seat, an HP Z8 or a Mac Studio Ultra, which is the "professional workstation" version of the same thing.

Table A3. A 2026 lab node against the A5K.

Axis	2026 lab node	A5K	Ratio
FP64 throughput	20 TFLOPS CPU + 270–320 TFLOPS GPU ≈ 300 TFLOPS	≈ 0 in the chipset; host does 10–50 GFLOPS	$\sim 10^4$–$10^5\times$
Memory	2–3 TB DDR5 + ~1 TB HBM	2 GB	$\sim 10^3\times$
Bandwidth	~1 TB/s DDR5 + 3–8 TB/s HBM	~13 GB/s	$\sim 10^2$–$10^3\times$
Storage	100+ TB NVMe	microSD	$\sim 10^5\times$
Cores	256 CPU + $\sim 10^5$ GPU lanes	1 RISC-V + fixed-function blocks	$\sim 10^5\times$
Software	NumPy, PETSc, FFTW, CUDA, MPI, every compiler	AROS, a demo scene, no BLAS	not comparable
Worst-case display latency	Unbounded: compositor, driver, scheduler	One scanline, by construction	A5K wins
Display-path power	5–30 W for the GPU alone, plus platform	1–2 W for the chipset	A5K wins
Lines of code to understand it	$\sim 10^8$	$\sim 10^4$	A5K wins

By how far: on the workload a mathematics lab actually runs — dense linear algebra, PDE solvers, FFTs, Monte Carlo — the workstation is **four to five orders of magnitude faster**, with a thousand times the memory and a software stack that has no counterpart at all. This is not close, and no amount of clever chipset design changes it.

Where it is not absurd: a laboratory also has *instruments*. Laser sync, ADC and DAC capture, camera triggers, optical benches, real-time control loops. There the question is not how many FLOPs but **when exactly a sample lands, and whether that can be proven**. A Linux box with a compositor and a fair scheduler gives statistical timing; this chipset gives a contract. That is a real, narrow, defensible niche — and precisely why §9.7 of the main document lists genlock and instrumentation rather than numerical work.

5. The honest bottom line

- **Versus AAA:** better by ~1.5 orders in silicon and 2–3 in memory and bandwidth, mainly because it is *buildable*. Worse in one respect: no native 68k.
- **Versus Hombre:** better in openness, cost, memory and survivability; worse in that Hombre had 3D and this does not.

- **Versus a 2026 lab workstation:** behind by 4–5 orders of magnitude on compute, 3 on memory, 2–3 on bandwidth, and effectively infinite on software. Ahead on exactly one axis — bounded display and timing latency — and on power and comprehensibility.

The truthful framing is not “a competitor to a modern PC”. It is: **the fastest, cheapest, most open way to own a machine whose display timing you can prove** — a 1994 architecture given 2026 memory and I/O, and a 1985 timing contract given a reason to still exist. At a mathematics lab it is not the computer that does the mathematics; at best it is the instrument that tells you, to the scanline, when the mathematics was displayed.

Table A4. The verdict at a glance.

Compared with	A5K ahead on	The other ahead on	Net
AAA (1994, †)	Buildability; ~10^2–10^3× memory and I/O; openness; 4K	Native 68k compatibility	A5K, by 1.5–3 orders
Hombre (1995, †)	Openness, cost, memory, survivability, tools	An integrated 3D pipeline	A5K, except in 3D
OCS / ECS / AGA (shipped)	~10^3× memory, 4K, PCIe, open toolchain	Shipped software, real 68k, a market	Different machines, not successors
2026 lab workstation	Bounded display latency, power, comprehensibility	10^4–10^5× FP64, 10^3× memory, the entire software stack	The workstation, by 4–5 orders

How these numbers were derived

- **OCS / ECS / AGA:** published hardware reference manuals and contemporary datasheets; figures are nominal, not measured on a specific board.
- **AAA and Hombre:** contemporary documentation and press coverage. Neither specification was ever frozen, so ranges are given rather than points, and the cancelling decisions mean no benchmark exists.
- **A5K:** the bill of materials in Appendix C of the main document, plus vendor datasheet figures for the ECP5-85 / Artix-7 200T families and DDR3L at 1600 MT/s.
- **Workstation:** current-generation vendor figures for a dual-socket, 128-core-per-socket node with eight accelerators. A single-seat workstation is roughly one order slower and half a terabyte smaller; the ratios do not change by more than that.
- **All ratios** are order-of-magnitude and rounded deliberately. A factor of two is noise at this scale; the arguments rest on orders, not decimals.

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